



EVERCARE

APPLICATION DEVELOPMENT

WEEK 4 ACTIVITY

CRUD Implementation

Journal Entries with Supabase persistence

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Selected entity

Journal Entries

EverCare caregivers can create, read, update, bookmark, filter, and safely delete private diary entries. Records are stored in Supabase PostgreSQL, while optional photos use a private Supabase Storage bucket.

4 CRUD OPERATIONS	SUPABASE DATABASE	OWNER-ONLY RLS
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Repository: github.com/Brenttttttt/evercare

How CRUD works in EverCare

The existing Journal Entries feature is the project entity used for Week 4. It satisfies the activity without adding duplicate demo-only logic: the production Flutter screens call one repository, and that repository performs owner-scoped operations against Supabase.

CREATE	READ	UPDATE	DELETE
New entry	List + reader	Edit entry	Confirm + remove

CRUD evidence matrix

Operation	User interface	Repository method	Persistent result
Create	Add Journal Entry + notebook form	createEntry()	INSERT into journal_entries
Read	Journal cards + full diary reader	fetchEntries()	SELECT owner rows, newest first
Update	Prefilled Edit Journal Entry form	updateEntry()	UPDATE row and reconcile photos
Delete	Destructive confirmation dialog	deleteEntry()	DELETE row and stored photos

Validation and user protection

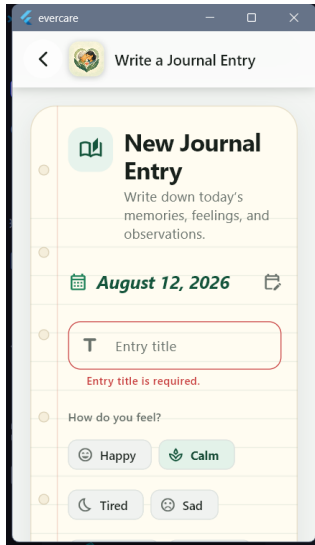
- Title and journal body are required and trimmed before saving.
- Custom tags are trimmed, limited to 30 characters, and checked for duplicates.
- The form prevents double submission and preserves writing after an error.
- Unsaved changes and permanent deletion both require confirmation.
- Photo attachments are limited to five and use clear loading/error states.

Persistence and privacy

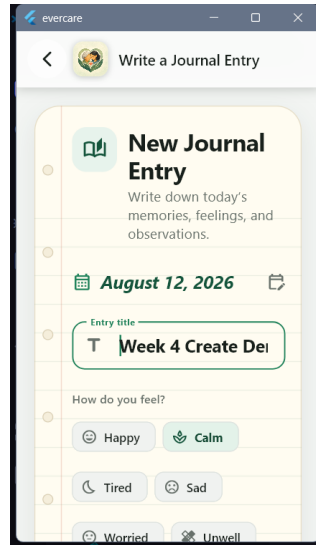
Supabase PostgreSQL stores journal records. Row Level Security restricts each query and mutation to `auth.uid()`. The repository also filters by `user_id`. Optional photos use the private journal-photos bucket, owner-scoped paths, and time-limited signed URLs. No service-role credential is embedded in Flutter.

Create operation and data validation

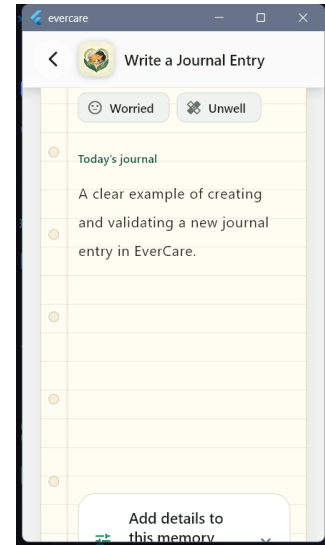
The create flow uses the production notebook-style form. The first screenshot shows required-field validation. The next screenshots show a completed title and journal body ready to save.

This screenshot shows the 'Write a Journal Entry' screen in the EverCare app. The title 'New Journal Entry' is at the top. Below it, the date is set to 'August 12, 2026'. The 'Entry title' field is empty and has a red border, with a red error message 'Entry title is required.' below it. The 'How do you feel?' section has four buttons: 'Happy', 'Calm', 'Tired', and 'Sad'. The 'Worried' and 'Unwell' buttons are at the bottom.

1. Required title validation

This screenshot shows the 'Write a Journal Entry' screen with a valid title. The 'Entry title' field now contains the text 'Week 4 Create De'. The 'How do you feel?' section has four buttons: 'Happy', 'Calm', 'Tired', and 'Sad'. The 'Worried' and 'Unwell' buttons are at the bottom.

2. Create form with a valid title

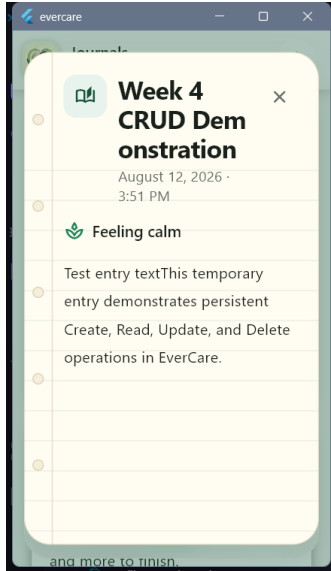
This screenshot shows the 'Write a Journal Entry' screen with the journal body entered. The 'Entry title' field contains 'Week 4 Create De'. The 'How do you feel?' section has four buttons: 'Happy', 'Calm', 'Tired', and 'Sad'. The 'Worried' and 'Unwell' buttons are at the bottom. A text area for the journal body contains the text 'Today's journal' and 'A clear example of creating and validating a new journal entry in EverCare.' A button at the bottom right says 'Add details to this memory'.

3. Journal body entered

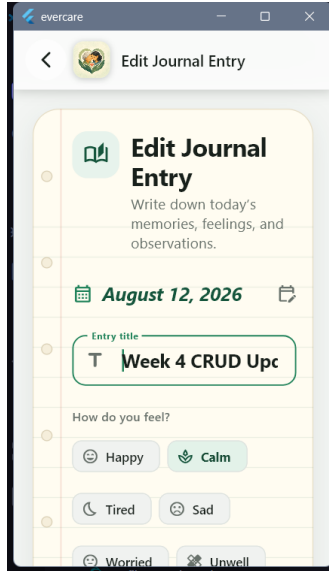
Create result: createEntry() inserts an owner-scoped journal_entries row in Supabase.

Read and update operations

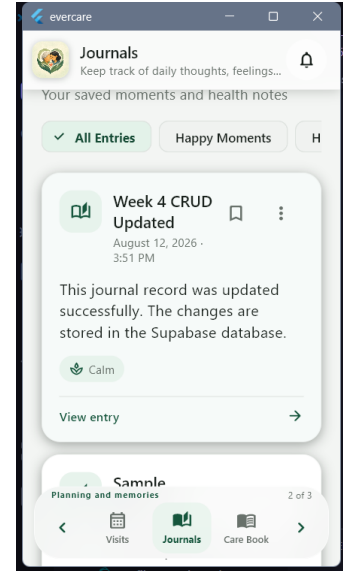
Saved entries are loaded from Supabase and displayed as journal cards. Selecting View opens the full diary reader. Editing restores the current values, then saves only the requested changes.



4. Read: full journal reader



5. Update: prefilled edit form



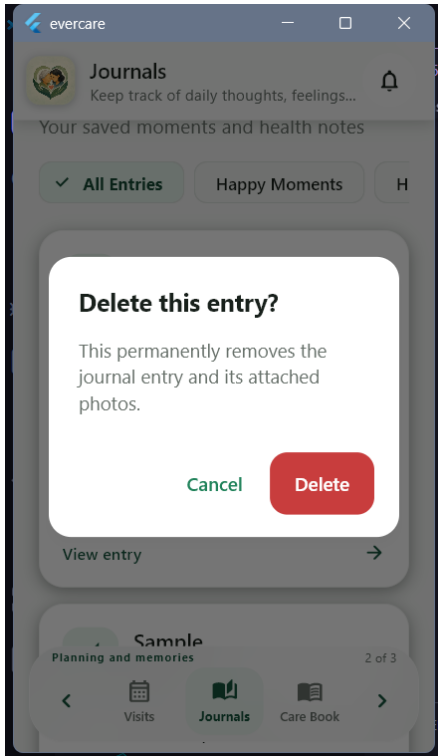
6. Updated record returned to the list

Update behavior

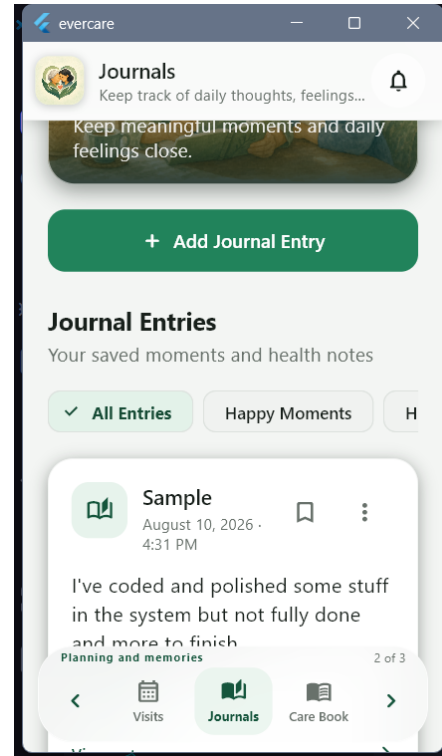
- Existing date, mood, title, body, symptoms, activities, tags, and photos are restored.
- The update query matches both the entry ID and the signed-in user ID.
- Unchanged photos are preserved; newly added and removed photos are reconciled.
- The list refreshes after a successful update, confirming the new persisted values.

Safe delete operation

Delete is intentionally destructive and requires explicit confirmation. The dialog explains that the journal entry and its attached photos will be permanently removed.



7. Permanent-delete confirmation



8. List refreshed after deletion

Deletion result

The temporary classroom record was removed after these screenshots were captured. `JournalRepository.deleteEntry()` first removes any owner-scoped `Storage` objects, then deletes the matching `journal_entries` row. Photo metadata is protected by owner RLS and cascades from the entry foreign key.

PRIVACY NOTE • Screenshots use non-sensitive classroom demonstration text. No patient health information was created for this submission.

Testing results

Check	Result	Evidence
Windows debug build	PASSED	Current source built successfully
flutter analyze	PASSED	No issues found
flutter test	PASSED	99 tests passed

Key implementation files

File	Responsibility
lib/repositories/journal_repository.dart	Supabase CRUD, photos, signed URLs
lib/screens/journals/journals_screen.dart	List, filters, actions, delete confirmation
lib/screens/journals/journal_entry_form.dart	Create/edit UI and validation
supabase/migrations/*.sql	Tables, constraints, private Storage, RLS

How to reproduce the CRUD workflow

1. Sign in to EverCare and open Journals.
2. Tap Add Journal Entry, complete the required title and body, then save.
3. Tap the card to read it; use the three-dot menu to edit or delete.
4. Confirm deletion and verify that the list refreshes without the record.

Submission checklist

- Updated Flutter project source code — included as EverCare_Week4_Source_Code.zip.
- Create, Read, Update, Delete, and validation screenshots — included in screenshots/.
- Brief CRUD documentation — this DOCX and matching PDF.
- Persistent Supabase database, owner-scoped RLS, and user-friendly confirmation flows.

GitHub repository

<https://github.com/Brenttttttt/evercare>